# 3rd-Party Libraries, Assets, Licensing & Implementation Status

## 1. 3rd-Party Libraries and Build Tools (UR_LIBRARY, QR_LICENCE)

| Library / Tool | Purpose in System | Licence | Suitability & Sustainability Analysis |
|---|---|---|---|
| libGDX (gdx, lwjgl3, box2d, freetype, tools) | Core game engine, 2D physics, and font rendering | Apache Licence 2.0 | Fully Suitable: Permissive licence granting modification, distribution, and patent rights without copyleft obligations. |
| Ashley, gdx-ai & Box2DLights | ECS architecture, AI utilities, and 2D lighting | Apache Licence 2.0 | Fully Suitable: Compatible with libGDX; permits closed-source or commercial resale by the customer. |
| LWJGL 3 (glfw, opengl, openal, stb, jemalloc) | Low-level native bindings for graphics and audio | BSD 3-Clause | Fully Suitable: Academic permissive licence allowing commercial redistribution if copyright notices are retained. |
| Construo (io.github.fourlastor) | Build plugin for packaging native executables | Apache Licence 2.0 | Fully Suitable: Build-time tool only; imposes no licensing conditions on distributed game binaries. |

## 2. 3rd-Party Assets (QR_LICENCE)

| Asset File | Source | Licence / Terms | Suitability & Sustainability Analysis |
|---|---|---|---|
| PixelifySans VariableFont_wght.ttf | Google Fonts | SIL Open Font Licence (OFL) 1.1 | Suitable: Open-source font licence permitting bundling, modification, and commercial distribution. |
| player.png | Avatars In Pixels (Minipix Maker) | Proprietary (Attribution; Non Commercial) | Prototype Only (Unsustainable): Acceptable for Assessment 1 academic evaluation, but violates commercial resale rules (QR_LICENCE). To be replaced with a CC0/custom sprite in Assessment 2. |
| bgd.png / bgd2.jpg (Grass textures) | iStock (Credit: Nastco) | Proprietary Stock (Paid Licence required) | Prototype Only (Unsustainable): Watermarked previews used temporarily. Will be replaced with CC0 or in-house textures in Assessment 2 to avoid paid dependencies (QR_LICENCE). |

## 3. Licence Compatibility & Overall Sustainability

In accordance with UR_LIBRARY and QR_LICENCE, all software dependencies use mutually compatible permissive licences (Apache 2.0 and BSD 3-Clause) free of copyleft restrictions (unlike GPL). This supports long-term maintainability (UR_MAINTAIN) for the team inheriting the codebase in Assessment 2 and enables future commercial distribution by the customer. Licensing risks are strictly isolated to the temporary third-party image assets (player.png, bgd.png, bgd2.jpg) decoupled inside assets/ and scheduled for replacement.

## 4. Unimplemented / Partially Implemented Assessment 1 Features

UR_INV / FR_INVINT, FR_ITEMUSE, FR_ITEMEFFECT, FR_ITEMCOUNT, FR_ITEMVALID (Inventory & Item Usage): The initial two inventory items and their dispersal effects on geese are not yet implemented in the Assessment 1 build.

UR_G_MOVE / FR_GEESE, FR_MOVESPEED (Geese Entities & Movement): Goose sprites (assets/Goose) exist in the repository, but active goose entities and chasing behaviour are not yet integrated into GameScreen.

UR_COUNTER / FR_COUNTERS (Usage & Event Counter): The on-screen counter tracking items used and hidden events triggered is partially implemented and pending UI integration.

UR_WINCOND, UR_RESULT / FR_WIN, FR_TIMEOUT, FR_RESULTSCREEN (Win/Loss & End Screen): While GameTimer tracks the 5-minute countdown (FR_TIMEDISPLAY) and LakeEvent/PitEvent implement hidden events (FR_EVENTHIDDEN), the game-over/victory state transition screen is not yet implemented.